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Integrated Speech Communication System with AI-Based Semantic Transmission

Stage 1 Progress Report: Conventional Digital Speech Communication System Baseline

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SUPERVISOR CERTIFICATION

I certify that the preparation of this project, entitled **“Integrated Speech Communication System with AI-Based Semantic Transmission (Stage 1 Report)”**, prepared by **Ghalia Qalajo and Hadeel Al-Shaar**, was carried out under my supervision at the Faculty of Engineering, Syrian Private University, in partial fulfillment of the requirements for the degree of B.Sc. in Intelligent Communications Engineering.

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ABSTRACT & KEYWORDS

Arabic Abstract (الملخص بالعربية)

تتطلب شبكات الاتصالات المتنقلة من الجيل السادس (6G) دعم تطبيقات متقدمة مثل الميتافيرس، الواقع الممتد (XR)، والإنترنت اللمسي، والتي تتطلب معدلات نقل بيانات عالية جداً وتأخيراً منخفضاً للغاية، بينما يظل الطيف اللاسلكي مورداً محدوداً. تقوم أنظمة الاتصالات الصوتية التقليدية بنقل شكل الموجة الصوتية الكاملة بت تلو الآخر، مما يستهلك حصة كبيرة من النطاق الترددي لنقل معلومات فائضة (Redundant). يقدم هذا المشروع تصميماً وتنفيذاً لنظام اتصالات صوتي متكامل. يركز هذا التقرير التفصيلي على المرحلة الأولى (Stage 1)، وهي بناء نظام اتصالات صوتي تقليدي (Conventional Speech Communication System) كمرجع أساسي (Baseline) للمقارنة، يتضمن ترميز المصدر، ترميز القناة، ونمذجة QAM/OFDM عبر قنوات الضجيج والقنوات المتلاشية (AWGN and Fading Channels)، وتقييم الأداء باستخدام الكفاءة الطيفية، معدل خطأ البت (BER)، الانتاجية، والتأخير الزمني.

الكلمات المفتاحية: الاتصالات الرقمية التقليدية، ترميز القناة، QAM، OFDM، الكفاءة الطيفية، قنوات التلاشي.

English Abstract

Sixth-generation (6G) mobile networks are expected to support applications such as the Metaverse, extended reality (XR), the Tactile Internet and holographic communication, all of which demand very high data rates and ultra-low latency, yet the radio spectrum remains a limited resource. Conventional speech communication systems transmit the full speech waveform bit by bit, so a large share of the bandwidth carries redundant information. This project designs and implements an integrated speech communication system that aims to raise spectral efficiency. This report presents Stage 1 of the project: a conventional speech communication system (source coding, channel coding, QAM/OFDM modulation) constructed as a baseline for comparison across AWGN and fading channels using spectral efficiency, latency, bit error rate, SNR, and throughput.

Keywords: Conventional Speech Communication, Baseline System, Source Coding, Channel Coding, QAM, OFDM, Spectral Efficiency, Fading Channels.

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List of Abbreviations and Symbols (فهرس المصطلحات والرموز)

Abbreviation	Meaning (المصطلح بالإنكليزية)	Arabic Meaning (المقابل بالعربية)
6G	Sixth-Generation Mobile Networks	شبكات الجيل السادس للمحمول
AI	Artificial Intelligence	الذكاء الصنعي
AMR	Adaptive Multi-Rate	معدل متعدد متكيف
AWGN	Additive White Gaussian Noise	ضجيج أبيض متناظر إضافي
BER	Bit Error Rate	معدل خطأ البت
BRNN	Bidirectional Recurrent Neural Network	شبكة عصبونية تكرارية ثنائية الاتجاه
CNN	Convolutional Neural Network	شبكة عصبونية التفافية
CP	Cyclic Prefix	البادئة الدورانية
CTC	Connectionist Temporal	تصنيف زمني اتصالي

	Classification	
FFT	Fast Fourier Transform	تحويل فوريه السريع
GAN	Generative Adversarial Network	شبكة توليدية تنافسية
GPU	Graphics Processing Unit	وحدة معالجة الرسوميات
LDPC	Low-Density Parity-Check	فحص التماثل منخفض الكثافة
LOS	Line of Sight	خط النظر المباشر
OFDM	Orthogonal Frequency-Division Multiplexing	التعدد التقسيمي بالتردد المتعامد
PAPR	Peak-to-Average Power Ratio	نسبة القدرة العظمى إلى المتوسطة
PCM	Pulse-Code Modulation	تعديل سعة النبضة
QAM	Quadrature Amplitude Modulation	تعديل سعة التعامد
SNR	Signal-to-Noise Ratio	نسبة الإشارة إلى الضجيج
STT	Speech-to-Text	تحويل الكلام إلى نص
TTS	Text-to-Speech	تحويل النص إلى كلام
XR	Extended Reality	الواقع الممتد

CHAPTER 1: INTRODUCTION

1.1 Overview & General Context

Mobile communication systems have evolved from voice-only networks to platforms that carry video, cloud services and, increasingly, immersive experiences. Each generation has pushed for higher data rates and lower delay, and 6G is expected to continue that trend while integrating artificial intelligence (AI) into the network itself, an idea often described as an AI-native network.

Classical communication theory treats the transmitter and receiver as a pipeline whose job is to deliver bits reliably, regardless of what those bits mean. The long-term vision of this overall project applies semantic concepts to a complete speech link, which gradually evaluates replacing raw speech waveforms with text and compact semantic representations. This report focuses specifically on Stage 1: building and fully evaluating a conventional digital speech communication chain as the reference baseline.

1.2 Problem Statement & Motivation

6G applications need very high data rates and ultra-low latency, while the radio spectrum is limited. Conventional speech systems send the full waveform, bit by bit. A large part of that traffic is redundant: it describes details of the signal that do not change the meaning of what was said. As the number of users and the demands of applications grow, spending bandwidth on redundant information becomes increasingly costly.

To quantify how much spectrum can be saved in future stages, an exact conventional baseline must first be implemented and tested. Table 1.1 outlines the key 6G applications that motivate this research.

Application	Why it is relevant to this project
Metaverse	Shared virtual worlds with many simultaneous users; voice is a core interaction channel and every saved bit helps scale.
VR / XR	Immersive headsets need high throughput and low delay for graphics; economical speech transmission leaves more capacity for other data.
Tactile Internet	Remote control with haptic feedback (for example remote robotic surgery) depends on

	very low latency over a network link.
Holographic calls	3D holographic video calls require very high data rates; audio should consume as little of the link as possible.

1.3 Research Objectives

The primary objective of Stage 1 is to design and implement a conventional speech communication system (source coding, channel coding, QAM/OFDM modulation) as a baseline for comparison. The specific objectives for Stage 1 include:

- Build a complete classical digital speech communication chain (PCM/source coding, Hamming/LDPC channel coding, QAM constellation mapping, and OFDM modulation).
- Model noisy and fading channel environments (AWGN, Rayleigh, and Rician fading) across a sweep of signal-to-noise ratios (SNR).
- Establish baseline reference values for spectral efficiency (η), end-to-end latency (T), Bit Error Rate (BER), and throughput (Th).
- Prepare a verified physical baseline dataset that serves as ground truth for evaluating future project enhancements.

1.4 Research Contributions

This stage provides a fully modular Python-based simulation environment that integrates speech audio processing with a classical digital transmission chain. It delivers verified mathematical equations, empirical simulation sweeps, and standard performance curves for classical voice communications over fading channels.

1.5 Beneficiary Entities

The beneficiaries include academic departments evaluating communication systems, wireless researchers modeling 5G/6G physical layer baselines, and engineering student teams requiring standardized physical link models.

1.6 Thesis Organization

The remainder of this report is organized as follows: Chapter 2 covers the theoretical fundamentals of traditional communication systems. Chapter 3 reviews three closely related literature studies [1]-[3]. Chapter 4 describes the methodology and system design for Stage 1. Chapter 5 presents evaluation setup and metrics. Chapter 6 summarizes current progress and future directions.

CHAPTER 2: THEORETICAL STUDY

2.1 Traditional Communication System Chain

A conventional communication system is a chain of blocks, mirrored at the receiver. At the transmitter, the source encoder compresses the speech, the channel encoder adds parity bits for error correction, and the modulator (QAM/OFDM) maps the bits to symbols that can be sent over the air. The signal then crosses the channel, where noise and fading are added. At the receiver, the demodulator, channel decoder and source decoder undo each step in reverse order to recover the speech.

2.2 Source Coding

Source coding compresses the signal by removing redundancy, so that fewer bits are needed to represent the same information. A classic example is Huffman coding, in which frequent symbols receive shorter codewords. Consider four symbols A, B, C and D with codewords A = 0, B = 10, C = 110 and D = 111. With probabilities (1/2, 1/4, 1/8, 1/8), the average length is:

$$\mathbf{L = 1\times(1/2) + 2\times(1/4) + 3\times(1/8) + 3\times(1/8) = 1.75 \text{ bits/symbol}}$$

compared with 2 bits per symbol for a fixed-length code. For audio, the same principle is realized by speech codecs such as PCM, AMR and Opus.

2.3 Channel Coding

Channel coding does the opposite of source coding: it adds controlled redundancy in the form of parity bits so that the receiver can detect and correct errors introduced by the channel. The ratio between the information bits k and the coded bits n is the code rate:

$$\mathbf{R_c = k / n}$$

In the Hamming (7,4) code, four data bits are expanded to a seven-bit codeword. For example, the data 1011 becomes the codeword 1011010 (the last three bits are parity). If noise flips a single bit, the decoder can locate and correct it. The code rate here is $R_c = 4/7$. Modern systems use convolutional, LDPC and Polar codes (the latter two are used in 5G).

2.4 Modulation: QAM and OFDM

Quadrature Amplitude Modulation (QAM) maps groups of bits to points on the I/Q plane, changing both the amplitude and the phase of the carrier. An M-QAM constellation carries log2(M) bits per symbol: 4-QAM (QPSK) carries 2 bits, 16-QAM carries 4 bits, 64-QAM carries 6 bits and 256-QAM carries 8 bits. A higher order gives a higher data rate, but the constellation points are closer together, so a higher SNR is needed to tell them apart.

Orthogonal Frequency-Division Multiplexing (OFDM) splits the data stream over many orthogonal subcarriers. At the transmitter an inverse FFT creates the signal, and at the receiver

an FFT recovers the subcarriers. The subcarrier spectra overlap, yet each peak falls at the zeros of the others, which is what makes the subcarriers orthogonal. OFDM is used in 4G, 5G and Wi-Fi.

Advantages of OFDM	Disadvantages of OFDM
Robust to multipath thanks to the cyclic prefix (CP)	High peak-to-average power ratio (PAPR)
High spectral efficiency, because subcarriers overlap	Sensitive to frequency offset and Doppler
Simple one-tap equalisation using the FFT	The cyclic prefix adds overhead

2.5 Wireless Channel Models

AWGN channel: The additive white Gaussian noise channel adds white, Gaussian-distributed noise to the signal. It is the ideal baseline channel where $y = x + n$.
Fading channel: In a wireless environment the signal reaches the receiver along several paths (multipath propagation), modeled by $y = h \cdot x + n$. When there is no line of sight (LOS) the gain h follows a Rayleigh distribution; when a line-of-sight path exists it follows a Rician distribution.

2.6 Evaluation Metrics

The following metrics are used to evaluate the Stage 1 baseline link performance:

Metric	Definition / Equation
Spectral efficiency (bit/s/Hz)	$\eta = R_b / B$ (Bit rate over occupied bandwidth; higher is better)
Latency (end-to-end delay)	$T = T_{src,enc} + T_{ch,enc} + T_{mod} + N_b/R_b + T_{prop} + T_{demod} + T_{ch,dec} + T_{src,dec}$
BER (bit error rate)	$BER = N_{error_bits} / N_{total_bits}$
SNR (signal-to-noise ratio)	$SNR = P_{signal} / P_{noise}$, $SNR_{dB} = 10 \cdot \log10(P_{signal} / P_{noise})$
Throughput (bit/s)	$Throughput = N_{correct_bits} / T \approx R_b \cdot (1 - BER)$

CHAPTER 3: LITERATURE REVIEW

3.1 Review of Related Studies

Three recent studies on speech semantic and conventional communications were reviewed as referenced literature [1], [2], [3]:

1. Zhou et al. (2024) [1] – 'Speech semantic communication based on Swin Transformer': Proposes DeepSC-TS (CNN + Swin Transformer) for speech semantic coding, achieving superior speech quality over AWGN and fading channels.

2. Weng et al. (2025) [2] – 'Robust semantic communications for speech transmission': Introduces Ross-S2T, using a deep encoder and GAN-based compensator to maintain high semantic similarity (>0.5 vs <0.4 for conventional) under low SNR Rayleigh fading.

3. Weng et al. (2023) [3] – 'Deep learning enabled semantic communications with speech recognition and synthesis': Proposes DeepSC-ST, utilizing STT (CNN+BRNN) and TTS (Tacotron 2) to transmit speech as text/semantic symbols, requiring ~12.7× fewer symbols than raw speech.

3.2 Comparative Analysis of Related Literature

Table 3.1 summarizes and compares the three reviewed literature studies.

Study	Approach	Main Reported Result
Zhou et al. (2024) [1]	CNN + Swin Transformer codec	Better speech quality than DeepSC-S baseline under identical load
Weng et al. (2025) [2]	Semantic encoder + GAN compensator	Semantic similarity > 0.5 vs < 0.4 at SNR = 1 dB (Rayleigh)
Weng et al. (2023) [3]	CNN + BRNN with CTC, then Tacotron 2	≈ 12.7× fewer symbols; readable text from −2 dB vs 4 dB SNR

3.3 Research Gap and Stage 1 Context

The literature demonstrates significant data reduction when transmitting compact representations. However, to quantitatively assess the exact performance gains of any novel

method, a fully implemented, standardized Stage 1 conventional digital link must be built and benchmarked across identical channel conditions. Study [3] serves as a primary reference for comparative analysis against classical STT/TTS chains.

CHAPTER 4: RESEARCH METHODOLOGY & SYSTEM ARCHITECTURE

4.1 Overview & System Architecture

Stage 1 establishes the baseline digital speech communication link. The transmitter compresses speech via source coding, applies forward error correction (channel coding), maps bits to QAM symbols, and modulates them using OFDM across the physical channel. At the receiver, equalisation, demodulation, channel decoding, and source decoding reconstruct the audio.

4.2 Stage 1 Baseline System Design

Stage 1 consists of the conventional digital speech chain:
Transmitter: Speech → Source coding → Channel coding + QAM modulation → OFDM IFFT
Channel: Audio bits over AWGN / Rayleigh fading channel
Receiver: OFDM FFT → Demodulation + Channel decoding → Source decoding → Speech

This stage fixes reference benchmark values for spectral efficiency, latency, BER, SNR, and throughput.

4.3 Evaluation Approach & Software Tools

Table 4.1 details the software tools used to construct and evaluate the Stage 1 baseline simulation.

Tool	Role in Stage 1 Project
Python	Main programming language for simulation and digital link implementation.
librosa	Audio loading, signal processing and feature extraction.
PyTorch / NumPy / SciPy	Matrix operations, signal processing, and physical link modeling.
Google Colab (GPU)	Hardware execution environment for simulation runs and processing.